

*Recent Research
on Radioactivity
by
Pierre Curie*

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RESEARCH ON RADIOACTIVITY ***

RECENT RESEARCH ON RADIOACTIVITY.

By P. CURIE.

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RECENT RESEARCH ON RADIOACTIVITY.^[1]

By P. Curie.

Since the discovery of strongly radioactive substances, research on radioactivity has been greatly developed. I propose in this article to give an account of the actual state of our knowledge relative to this subject, laying particular stress on the most recent work.^[2]

^[1]Translated from an article that appeared in *Jour. d. Chim. Phys.*, I, 409 (1903) edited by Philippe A. Guye, Professor of Chemistry in the University of Geneva. See THIS JOURNAL, 31, 298 (1904).

^[2]For more complete details of work done previous to May, 1903, see the thesis of Mme. Curie. It appeared in *Ann. de Chim. et de Phys.* in 1903 and 1904.

I. Radioactive Substances.

Becquerel Rays. Uranium and Thorium.—We call radioactive such substances as are capable of emitting spontaneously and continuously certain rays known as Becquerel rays. These rays act upon the photographic plate; they render the gases through which they pass conductors of electricity; they can pass through black paper and metals. The Becquerel rays cannot be reflected, refracted or polarized.

In 1896 Becquerel discovered that uranium and its compounds emit these new rays continuously. Schmidt and Mme. Curie then found almost simultaneously that thorium compounds are also radioactive. The radiations emitted by thorium compounds are comparable in intensity with those from the compounds of uranium. Radioactivity is an atomic property that accompanies the atoms of uranium and thorium wherever they are found; in a compound or a mixture its intensity is proportional to the amount of the metal present.

New Radioactive Substances.—Mme. Curie, in 1898, tried to learn whether there were among the elements then known any others possessing radioactivity; she could not find a single substance giving any considerable radiation, and concluded that the radioactive properties of the elements are at least 100 times more feeble than those of uranium and thorium. She found, on the other hand, that certain minerals containing uranium (pitchblende, chalcocite, and carnotite) are more active than metallic uranium; the activity of these minerals could not, then, be due solely to uranium or to other known elements. This discovery was fertile with new results. Mme. Curie and I showed, in an investigation carried on together, that pitchblende contains new radioactive substances, and we supposed that these substances contained new chemical elements.

There are known with certainty three new strongly radioactive substances: polonium, which was found in the bismuth obtained from the uranium minerals; radium, found with barium from the same source, and actinium, which was discovered by Debierne among the rare earths extracted from the same minerals. These three substances are present only in infinitesimal quantities in the uranium minerals, and all three possess a radioactivity about a million times greater than that of uranium or thorium.

Recently Giesel and Hofmann announced the presence of a fourth strongly radioactive substance in the uranium minerals, which had

properties closely resembling those of lead; from the publications that have appeared up to this time I have not been able to form an opinion as to the nature of this substance.

It may be asked whether radioactivity is a general property of matter. This question cannot be regarded as actually settled. The investigations of Mme. Curie have proved that the different known substances do not possess an atomic radioactivity one-hundredth as great as that of uranium or thorium. On the other hand, certain chemical reactions may cause the formation of ions, conductors of electricity, without the active substance giving any evidence of atomic radioactivity. Thus white phosphorus by its oxidation renders the surrounding air a conductor of electricity, while red phosphorus and the phosphates are not at all radioactive.

Some old experiments by Russell, Colson and Tengyel showed that certain substances act upon the photographic plate at a distance. It is possible that this phenomenon is partly due to radioactivity, but it is not certain. Recent work by MacTennan and Burton, Strutt, and Lester Cooke, leads to the supposition that radioactivity belongs to all bodies to an extremely slight extent. The identity of these very feeble phenomena with atomic radioactivity can not yet be regarded as proved.

Radium.—Of all the strongly radioactive substances radium is the only one that has been proved to be a new element. Radium possesses a characteristic spectrum, the discovery and first study of which we owe to Demarçay, and which has since been studied by Runge and Precht and by Crookes. Radium is an element belonging after barium in the series of alkaline earths. Its atomic weight as determined by Mme. Curie is 225.

At present radium is obtained from a residue left on extracting uranium from pitchblende. This residue contains 2 or 3 decigrams of radium in a ton. At first 10 or 15 kilograms of radiferous barium salt are extracted from a ton of the residue, and from this the radium is prepared by fractional crystallization (with the chloride or bromide), the crystals that form from the solution being richer in radium than the salt that remains in the liquid.

The radiant activity of the radium salt can be measured at different times from the beginning of crystallization until it is finally dried in the air-bath. It is found that the activity has a certain initial value, then it increases as a function of the time, at first rapidly, then more and more slowly; it

approaches asymptotically a limiting value which is about five times as great as the initial activity. The activity then remains invariable for some years if the salt be left as it is.

Polonium.—Polonium is, on the contrary, a substance that slowly loses its radioactivity from the moment when it is separated from the uranium mineral containing it. After some years the radioactivity of polonium almost completely disappears. Hence it acts like an unstable substance. It has not yet been shown that polonium is a new element distinct from ordinary bismuth.

Polonium can be concentrated by fractionally precipitating the subnitrate of polonium and bismuth by means of water. The part precipitated is the most active. One can also make a partial precipitation by means of hydrogen sulphide from a strongly acid solution in hydrochloric acid. These methods of fractionation are difficult because the precipitates are not easily redissolved. Marckwald concentrated the activity by plunging a rod of bismuth in a solution of bismuth and polonium. A layer of extremely active metal was deposited upon the rod.

Actinium.—The concentration of actinium is even more difficult than that of polonium. Solid salts containing actinium preserve their radioactivity unchanged for several years.

II. The Radiations from Radioactive Substances.

Complexity of the Radiation.—Radium is the substance whose radioactivity has been studied most completely. It is now known that it emits a number of rays of different nature, which can be placed in three groups. According to the notation adopted by Rutherford, I designate the three groups of rays by the letters α , β , and γ .

The action of the magnetic field serves to distinguish them. In an intense magnetic field the α -rays are slightly deviated from a straight path, and the deviation is of the same kind as that of the "canal" rays of Goldstein in a vacuum-tube. On the other hand, the β -rays are deviated like the cathode rays, and the γ -rays are not turned aside, but act like the Roentgen rays.

β-Rays.—The β -rays of radium, analogous to the cathode rays, form a heterogeneous group. They are distinguished from one another by their power of penetration and by the deviation caused by a magnetic field.

Certain β -rays are absorbed by aluminium foil a few hundredths of a millimeter in thickness, while others pass through several millimeters of lead.

Suppose we have a rectilinear pencil of Becquerel rays obtained by means of a particle of radium salt and a screen with a hole in it. If a uniform magnetic field is produced normal to the direction of the pencil the β -rays are incurved, and describe circular paths in a plane normal to the direction of the magnetic field. The rays of the described circumferences vary within wide limits. Becquerel has shown that the most penetrating rays are least deviated, and hence describe circumferences with the greatest radius of curvature. If the pencil of β -rays bent aside by the magnetic field is allowed to fall on the photographic plate, there is obtained an impression that is a true spectrum in which the different β -rays act separately.

Suppose the β -rays to be projectiles (electrons) negatively charged with electricity and shot from the radium with great velocity. Let m be the mass of a projectile, e its charge, v its initial velocity, r the radius of curvature of its path, H the intensity of the magnetic field normal to the initial velocity, μ the magnetic permeability of the medium. We then have the easily established relation

The β -rays are also deviated in an electric field. Suppose we have a rectilinear pencil of these rays. If a uniform electric field be produced normal to the initial direction of the pencil, the rays are deviated in the opposite direction to the field, and describe parabolic paths. The experiment can be performed by passing the pencil of rays between two parallel metallic plates, between which there is established a difference of potential. The deviation is slight with the means one has at command, and it is convenient to work in a vacuum. The air is made a conductor of electricity,

so that if we work in the air the insulation is imperfect, and it is difficult to maintain a constant and high difference of potential between the plates. The most penetrating β -rays are bent aside least.

The action of the electric field is in accord with the ballistic hypothesis previously stated. Accepting this hypothesis, suppose a uniform electric field with an intensity E and a width l , acts upon the charged projectile whose initial velocity is normal to the field. The deviation, d , of the extremity of the trajectory at the departure from the field is given by formula (2), when the deviation is slight.

From the equations (1) and (2) can be deduced the velocity, v , of the projectiles, and the ratio, e/m , of the electric charge to the corresponding mass.

The experiments of Becquerel have shown that for the β -rays of greatest intensity the ratio, e/m , approaches 10^7 electromagnetic units, and v a value of 1.6×10^{10} cm/sec. These values are of the same order of magnitude as for the cathode rays.

Kaufmann has made exact experiments on the same subject. This physicist subjected a very narrow pencil of radium rays to the simultaneous action of a magnetic and an electric field, both fields being uniform and acting normal to the primary direction of the pencil. The pencil is received upon a photographic plate placed perpendicular to its initial direction. In the absence of the two fields the impression on the plate is a small circular disk, almost like a point. When the magnetic field acts alone the different β -rays are unequally deflected but remain in a plane normal to the field, producing upon the plate an impression in the form of a straight line. When the electric field acts alone the different β -rays are unequally deflected in a plane passing through the field and producing upon the plate an impression which is a straight line perpendicular to the one preceding. When both fields act simultaneously the impression on the plate is a curve. Each point of the curve corresponds to a different kind of β -ray. Taking as axes of coordinates

on the plate the straight lines obtained when each of the two fields acts alone, the coordinates of each point of the curve represent the relative magnetic and electric deviations for each kind of ray.

The following are the figures for e/m and v obtained by Kaufmann, referred to the most penetrating rays of radium. I give for comparison the values obtained by Simon for the cathode rays.

e/m in electromagnetic units.	v cm/sec
1.865×10^7	0.7×10^{10} (For the cathode rays. Simon.)
1.31×10^7	2.36×10^{10}
1.17×10^7	2.48×10^{10}
0.97×10^7	2.59×10^{10} (For the radium rays. Kaufmann.)
0.77×10^7	2.72×10^{10}
0.63×10^7	2.83×10^{10}

It is seen that certain β -rays have a velocity approaching that of light. One can understand why it is that particles having such velocity should be able to penetrate matter to such an extent, although they are so small.

The ratio e/m , seems to be the same for the least penetrating radium rays and the cathode rays. But this ratio diminishes as the velocity of the rays increases. J. J. Thomson and Townsend think that all charged electrons, when in motion, possess the same charge, which is equal to that carried by an atom of hydrogen when a solution is electrolyzed. If that be true, it must be admitted that the mass of the particles increases with their velocity, when the latter approaches that of light.

In the case of electrolysis the ratio, e/m , is equal to 9650, while this same ratio is equal to 1.865×10^7 for the cathode rays and the less penetrating β -rays. If it be admitted that the charge, e , is the same in both

cases, it can be deduced that the mass of an electron is about two thousand times less than that of an atom of hydrogen.

Theoretical considerations lead to the conception that the inertia of a particle is solely due to its being a moving charge, the velocity of a moving electric charge not being altered without some change of energy. In other words, the mass of the charged particle is, in part at least, an apparent mass or an electromagnetic mass. Abraham gave a formula for calculating the electromagnetic mass of a charged particle as a function of its velocity. According to this formula the mass due to electromagnetic reactions is a constant for low velocities; it increases with the velocity, and approaches infinity for velocities that approach that of light. The experiments of Kaufmann are in accord with this theory, and also lead to the conclusion that the mass of an electron is entirely of an electromagnetic nature. These results are of great theoretical importance. One can foresee the possibility of establishing mechanics upon the dynamics of small material centers charged and in motion.

α -Rays.—The α -rays of radium have very little power of penetration. A piece of aluminium foil a few hundredths of a millimeter in thickness absorbs them almost completely. They are also absorbed by the air, and cannot penetrate air at the atmospheric pressure to a greater distance than 10 cm. The α -rays form the most important part of the radiation of radium, provided we measure the radiation by the amount of ionization which it produces in the air.

The α -rays are very slightly deflected by the most intense magnetic and electric fields, and they were at first thought to be non-deviable under this influence. Nevertheless, independently of the action of the magnetic field, the laws of the absorption of the α -rays by superimposed screens is sufficient to distinguish them clearly from the Roentgen rays. In passing through successive screens the rays become less and less penetrating, while the penetrating power of the Roentgen rays increases. The ray is like a projectile whose energy diminishes on passing through each screen. A given screen also absorbs the α -rays to a much greater extent when it is placed at a distance than when it is placed quite near the radium.

Strutt suggested that the rays might be analogous to the canal rays of vacuum-tubes. Rutherford succeeded in showing the action of a magnetic

field upon the rays and in making a preliminary measurement of the deviation. Becquerel confirmed the results obtained by Rutherford and gave a new measurement of the phenomenon. Des Coudres measured the electrical and magnetic deviation of the rays in a vacuum.

From these investigations it is seen that the α -rays act like projectiles having a great velocity and charged with electricity. The deviation in a magnetic field and in an electric field is the opposite of the deviation of the cathode rays.

The α -rays form an apparently homogeneous group, all being deflected to the same extent and, hence, not giving a spread-out spectrum as the rays do. The formulas (1) and (2) are applicable. According to the measurements of Des Coudres in a vacuum,

Hence the velocity of the particles is twenty times less than that of light. If we assume that the charge of a particle is the same as that of an atom of hydrogen, in electrolysis, it is found that its mass is of the same order of magnitude as the hydrogen atom (the ratio, m/e , is equal to 9650 for hydrogen in electrolysis). It may well be supposed that these particles, which are greater than the electrons and have a lower velocity, would be less capable of penetration.

According to the experiments of Becquerel, the curvature of the path of the α -rays moving in a uniform magnetic field is not constant when in air at the atmospheric pressure. At first the curvature is the same as that obtained in a vacuum, but it becomes less and less as the ray recedes from its source. This phenomenon can be explained by supposing that new particles attach themselves to the projectiles that make up the rays while the latter move through the air. This hypothesis explains why the absorbing power of a screen for the α -rays increases as it is removed farther from the radiant source.

The α -rays are the ones that are active in the very beautiful experiments with the spinthariscopes of Crookes. In this apparatus a very small fragment

of a salt of radium (a fraction of a milligram) is held by a metallic thread a short distance (0.5 mm.) from a phosphorescent screen of zinc sulphide. On examining, in the dark, the surface of the screen which is toward the radium by means of a magnifying glass, luminous points are seen sprinkled over the screen, reminding one of a starry sky. These luminous points appear and disappear continuously. From the ballistic theory it might be imagined that each luminous point that appeared and disappeared resulted from the shock of a projectile. This is the first instance of a phenomenon which enables one to distinguish the individual action of an atom.

γ-Rays.—The rays are exactly like the Roentgen rays. They seem to comprise but a small part of the total radiation. The γ -rays have a very extraordinary power of penetration, and they diffuse very little in passing through most substances.

Diffusion of the Radium Rays.—Suppose we have a pencil of Becquerel rays issuing from radium and limited by a screen of lead. If the pencil meets a thin screen the α -rays are absorbed, the β -rays are diffused in all directions, and the γ -rays go part way through the screen as a well-defined pencil with sharp edges. The γ -rays can also pass through a prism of thick glass as a well-defined straight pencil. The question has been raised whether the β -rays are always completely diffused when they penetrate a solid screen. The experiments of Becquerel show that a pencil of β -rays can propagate itself in a well defined way in paraffin. Becquerel made use of the action of the β -rays on the photographic plate to study the path of the rays when dispersed by a magnetic field. It can be seen from the prints that the most penetrating rays pass through 7 or 8 mm. of paraffin without marked diffusion, while the least penetrating rays are completely diffused after passing through 2 mm. The magnetic field deflects the β -rays in paraffin as in air.

Conductivity of Dielectric Liquids under the Influence of the Radium Rays.—Dielectric liquids become poor conductors under the influence of the radium rays. This can be shown with petroleum, ether, vaseline oil, benzene, amylene, carbon disulphide, and liquid air.

Radiation of Other Radioactive Substances.—Polonium emits only very slightly penetrating rays, which seem to be identical with the α -rays of radium. They possess almost the same power of penetration and are

deflected in the same way by the magnetic field. Finally, the experiment of the spinthariscopes can be performed with the α -rays of polonium. Hence, polonium is a source of the α -rays unmixed with other kinds of rays, which is of value in certain studies. But the source exhausts itself, and some years after it has been separated from the minerals containing it the polonium loses its activity.

Thorium, uranium, and actinium seem to emit the α - and β -rays; the deviability of the β -rays has been shown.

The Electric Charge of the Radium Rays.—According to the ballistic theory, the α -rays should carry positive electric charges and the β -rays negative electric charges. Mme. Curie and I have shown that, in conformity to the theory, the β -rays of radium charge positively the bodies that absorb them. To show this, a plate of lead is connected with an electrometer. The plate of lead is entirely covered with a layer of paraffin, which is in turn enveloped in fine aluminium foil connected with the earth. The radium, placed in a small dish, sends its rays upon the plate of lead, which is thus protected. The α -rays are stopped by the exterior layer of aluminium. Part of the β -rays pass through the aluminium and the paraffin and are absorbed by the lead, which becomes charged negatively. The paraffin is necessary to sufficiently insulate the plate of lead, which could be charged if it were surrounded by air, for it is made a conductor by Becquerel rays.

We have also shown that a salt of radium is charged positively when it is enveloped in an insulating layer, and that it emits β -rays from the exterior, while the α -rays cannot escape.

A sealed glass bulb containing a salt of radium becomes spontaneously charged like a Leyden jar. If after a sufficient time a mark is made on the wall of the flask with a glass-cutter, a spark is discharged, which pierces the glass at the point where it was made thin by the cutter. At the same time the experimenter feels a slight shock in his fingers from the passage of the discharge.

Phosphorescence of Substances under the Action of the Becquerel Rays. Light Emitted by Salts of Radium. Coloration of Substances by the Action of the Rays.—The radiation from radium causes phosphorescence in a great many substances: the salts of the alkalies and alkaline earths, uranyl-potassium sulphate, organic substances, cotton, paper, cinchonine sulphate,

skin, glass, quartz, etc. The most sensitive substances are barium platinocyanide, willemite (silicate of zinc), sulphide of zinc, and diamond. With the penetrating β -rays willemite and the platinocyanide are the most sensitive, while with the α -rays it is better to use phosphorescent sulphide of zinc.

Phosphorescent substances are altered by the prolonged action of the radium rays. They become less excitable and are less luminous under the influence of the rays. At the same time they change their color or become colored. Glass becomes violet, black or brown. Salts of the alkalies turn yellow, green or blue. Transparent quartz becomes smoky quartz. Colorless topaz turns yellow, orange, etc. Glass colored by radium is thermoluminescent. On heating it to 500° it emits light. At the same time it becomes colorless and returns to its original condition. It is then capable of being colored anew by the action of the rays of radium.

The salts of radium are spontaneously luminous. It might be said that they make themselves luminous by the action of the Becquerel rays they emit. Anhydrous chloride and bromide of radium are the salts that give the most intense luminescence. They may be obtained so luminous that the light can be seen in full daylight. The light emitted by the salts of radium recalls in tint that from a fire-fly. The luminosity of the radium salt decreases with time without ever completely disappearing, and those that were colorless at first become gray, yellow or violet.

Physiological Effects of the Radium Rays.—The rays from radium cause different physiological actions.

A salt of radium placed in an opaque box made of cardboard or metal acts on the eye and produces the sensation of light. To obtain this result, the box containing the radium is placed before the closed eye or against the temple. In this experiment the center of the eye becomes luminous by phosphorescence under the influence of the radium rays, and the light one sees has its source in the eye itself.

The rays of radium act on the epidermis. If we place on the skin for a few minutes a bulb containing radium no particular sensation is felt. But fifteen or twenty days afterward it produces a reddening of the skin, then a slough in the place where the bulb was applied. If the action of the rays be long enough there is finally formed a sore that takes several months to heal.

The action of the rays from radium is analogous to that produced by the Roentgen rays. The attempt has been made to utilize this action in the treatment of lupus and cancer.

The radium rays also act upon the nervous centers and cause paralysis and death. (Danysz.) They seem also to act with especial intensity upon growing tissues. (Bohn.)

The Employment of Radium in the Study of Atmospheric Electricity.—The radium rays have also been utilized in the study of atmospheric electricity (Paulsen, Witkowski, Moureaux). A small quantity of a salt of radium placed at the extremity of a metallic rod serves as a point of contact for the potential. By this very simple arrangement we can avoid the use of flames or of water-dropping apparatus to measure the potential at any point in the atmosphere.

III. The Heat Given Off by the Salts of Radium.

The salts of radium continually give off heat. Its amount is sufficient for it to be detected by means of a crude experiment with two ordinary mercury thermometers. Two identical heat-insulating vacuum-bulbs are used. In one of these is placed a glass bottle containing 0.7 gram of pure radium bromide; in the second is placed a glass bottle containing some inactive substance, such as barium chloride. The temperature of the contents of each is shown by a thermometer placed with its bulb near the bottle. The mouths of the flasks are closed with cotton. Under these conditions the thermometer in the same flask as the radium always has a temperature 3° higher than that indicated by the other thermometer.

The quantity of heat given off can be estimated by means of a Bunsen ice calorimeter. By placing in the calorimeter a glass bottle containing a salt of radium, it is found that there is a continued supply of heat that stops as soon as the radium is removed. A determination made with a salt of radium that had been prepared a long time before shows that each gram of radium gives off 80 small calories an hour. Hence radium gives off enough heat to melt its own weight of ice every hour. Nevertheless the radium salt seems to

remain in the same condition, and besides, not a single ordinary chemical reaction can be called in to explain such a continuous liberation of heat.

It has been shown, also, that a recently prepared salt of radium sets free a relatively small amount of heat. The heat set free in a given time then increases continually, and tends towards a definite value which is not quite reached at the end of a month.

When a salt of radium is dissolved in water and the solution enclosed in a sealed tube, the amount of heat given off at first is slight. It then increases and tends to become constant at the end of a month. When the limit is reached, the amount of heat given off from the radium in solution is the same as if it were in the solid state.

The amount of heat set free by radium at different temperatures can also be estimated by causing it to boil a liquefied gas and measuring the volume of gas that is liberated. The experiment can be performed with methyl chloride, which boils at -21° .

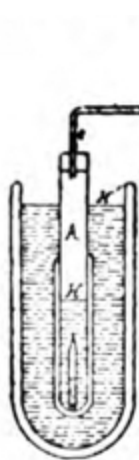


Figure I

The experiment was also performed by Dewar and myself with liquid oxygen (-180°) and liquid hydrogen (-252°). The latter is especially suitable for the experiment. A tube, *A*, closed at the lower end and surrounded by a Dewar vacuum-bulb, contains a little liquid hydrogen. A delivery-tube, *B*, allows the gas to be collected in a graduated tube filled with water. The tube, *B*, and its insulator are plunged into a bath of liquid hydrogen, *C*. Under these conditions no gas is produced in the tube *A*. When a bottle containing 0.7 gram of radium bromide is placed in the hydrogen in *A*, there is a continuous liberation of hydrogen gas, and 73 cc. of the gas are collected per minute. (The radium bromide had been made only ten days before.)

IV. Induced Radioactivity and Radioactive Emanations.

Induced Radioactivity.—Radium, thorium, and actinium have the property of acting externally, apart from the Becquerel rays that they emit. They communicate little by little their radioactive properties to substances in their neighborhood, and the latter emit in turn Becquerel rays. The activity can be transmitted to gases, liquids, and solids. The phenomenon is known as induced radioactivity.

Induced radioactivity propagates itself in gases step by step by a sort of conduction, and is not at all due to direct radiation from the body which causes it.

When the activated substance is removed from the radioactive body, the induced radioactivity on it persists for a certain time. It diminishes, nevertheless, little by little and finally disappears.

Emanation.—To explain these phenomena, Rutherford supposes that radium and thorium are constantly giving off an unstable, radioactive, gaseous material, which he calls an emanation. The emanation spreads into the gas surrounding the radioactive substance. It destroys itself little by little by giving off Becquerel rays and by producing other unstable radioactive substances that are not volatile. These new substances attach themselves to the surface of solid bodies and render them radioactive.

Without stating so many hypotheses, we may adopt the name emanation to designate the radioactive energy in the form which it has when it spreads into the gas surrounding the radioactive substance. It may be supposed also, that this energy disappears in creating the energy of induced radioactivity in solids.

The Radioactivity Induced by Radium and the Emanation from Radium.—When a solid salt of radium is placed in a closed space filled with air, the interior walls of the closed space and all solids in it become radioactive. We may, for example, place inside the space a plate of any solid, leave it there a certain time, and then remove it to study its activity. It is found that the activity of the plate increases at first according to the length of its stay in the space, but that it reaches a limiting value after a sufficiently long time. When the excited plate is removed from the space it loses its activity according to an exponential law, the radiation falling to one-half its value in about half an hour. In a general way all solid substances under the same conditions acquire and lose their activity in the same manner.

The phenomena are much more intense (about twenty times), if instead of the solid salt of radium, we place a solution of it inside the closed space in an open vessel.

The nature and the pressure of the gas contained in the closed space have no influence upon the observed phenomena.

The induced activity in a closed space is proportional to the quantity of radium present.

When the space containing the radium communicates with a second by a tube, solids contained in the second become equally active after a sufficient time. The transmission of the activating property can even take place from one to the other through a capillary tube.

When the gas which has been made active by remaining in one space containing radium is passed into another, it retains for a considerable time the property of rendering radioactive solids that are brought in contact with it. The gas thus removed from the action of the radium gradually loses, nevertheless, its power of causing radioactivity. It disappears as a function of the time, according to an exponential law. It loses one-half of its value every four days.

To interpret this phenomenon it must be assumed that radium continuously gives off a constant radioactive emanation. This emanation spreads into the air in a closed space, and acting on solids, makes them radioactive. When the air is transferred to another space, the emanation is carried along with it. Finally, it is destroyed spontaneously with such velocity that the quantity of emanation in the gas diminishes by one-half every four days.

In a space containing radium there is established a state of equilibrium when the quantity of emanation in the space is such that the loss of emanation resulting from its spontaneous destruction is exactly compensated by the continued supply coming from the radium.

The following experiment can be performed: The glass receiver, (Fig. II.), filled with air, communicates through the constricted part, , with the bulb, , which contains radium, . At the end of a certain time the emanation has spread into , and its interior walls have become active. The receiver, , is then separated from the radium by closing with a flame.

The external radiation of may then be studied by placing it in the interior cylinder of a cylindrical condenser (Fig. III.). The interior cylinder is of aluminium, . It is brought to a potential of 500 volts. The external cylinder, , of the condenser is of copper. It is in connection with an electrometer and piezoelectric quartz. By means of the quartz the current passing through the condenser is measured. This current is caused by the Becquerel rays that escape from the tube , pass through the cylinder of aluminium , and make the air between the two cylinders a conductor of electricity. The apparatus is surrounded by a protective metallic envelope, , connected with the earth.

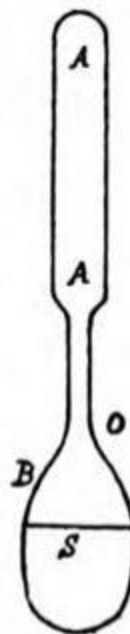


Figure II

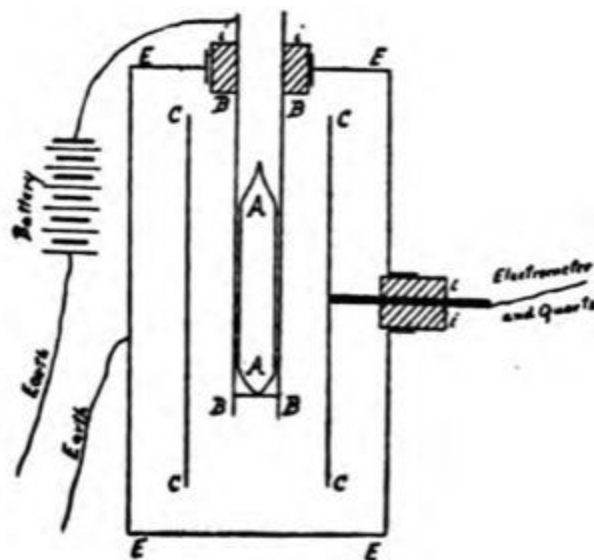


Figure III

It is found that the radiation from the tube diminishes with time according to an exact exponential law, expressed by the equation

R_0 is the value of the radiation at the beginning of the time, R the value at the instant t , and k a constant coefficient, $k = 2.01 \times 10^{-6}$, taking one second

as the unit of time. The radiation falls to one-half its value after about four days.

In a second experiment the tube may be made active as before, and the interior can then be evacuated by pumping out the air containing the emanation. Under these conditions the radiation of the receiver, , diminishes much more rapidly, and becomes twice as feeble in about half an hour. This law of loss of activity is the same as that according to which excited bodies lose their activity when they are exposed to the free air. The result is the same if inactive air be admitted to the receiver, , after having been evacuated.

One is thus led to the conclusion that in the first experiment the activity of the receiver, , is caused by the air charged with the emanation contained in the receiver, and that the law of the diminution of the radiation in this experiment represents as well the law of the spontaneous disappearance of the emanation.

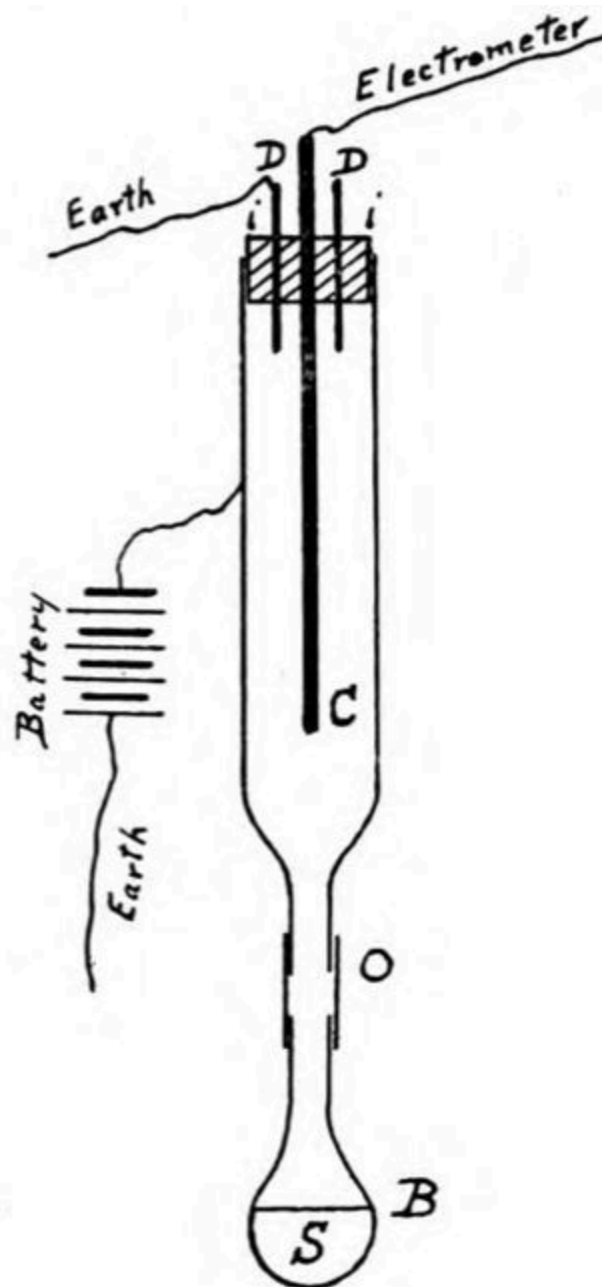


Figure IV

When a vacuum is produced in the receiver, , which contains air charged with the emanation, and when the radiation of the receiver is measured immediately after the extraction of the air, it is found that the radiation has not changed at the moment when the active air is withdrawn. The Becquerel radiation of air charged with the emanation does not, then, produce any action in this experiment. This radiation probably exists, but it

is composed of very slightly penetrating rays, incapable of passing through the glass wall. In this connection the following experiment can be performed. One of the ends of the metallic tube, (Fig. IV.), communicates at by means of a rubber tube with a receiver, , in which is a solution of a radium salt. The other end of the tube, , is closed with an insulating stopper, . Through the stopper passes a metallic rod, , connected with an electrometer. The tube, , and the rod, , form a cylindrical condenser. The tube, , is brought to a potential of 500 volts. The metallic tube, , connected with the earth, serves as a guard-tube. When the tube, , is sufficiently active it is removed from the radium, and the intensity of the current passing through the condenser is measured. Then the active air which fills the condenser is rapidly driven out, inactive air is admitted, and a new measurement of the intensity of the current is made immediately. It is found that the current has become six times more feeble. Thus, during the second measurement the radiation of the excited walls acts solely to ionize the air in the condenser, while during the first measurement the emanation acts as well. We may, then, suppose that it also emits an emanation. This radiation is necessarily very slightly penetrating, for its action cannot be detected on the exterior.

When a solid plate which has been excited by the emanation becomes inactive in free air, the law of loss of activity depends on the time during which the plate has been left in contact with the emanation. If the action of the emanation is prolonged (more than twenty-four hours, for example), the law of loss of activity is given by the difference between two exponentials. The intensity of the radiation, , may in this case be represented as a function of the time, , by the equation

I_0 is the intensity of the radiation at the beginning of the time, that is at the moment when the plate is removed from the influence of the emanation. , , and are three constant coefficients: $a = 4.2$; $b = 0.000413$; $c = 0.000538$, taking one second as the unit of time.

These results are represented by the curve 1 (Fig. V.). The logarithm of i is represented on axis of the ordinates, and the time on the axis of abscissas. One hour and a half after the beginning of the loss of activity the second exponential has become negligible with respect to the first in the expression for the value of i , and the representative curve has become straight. From this moment the activity diminishes by one-half during each period of twenty-eight minutes.

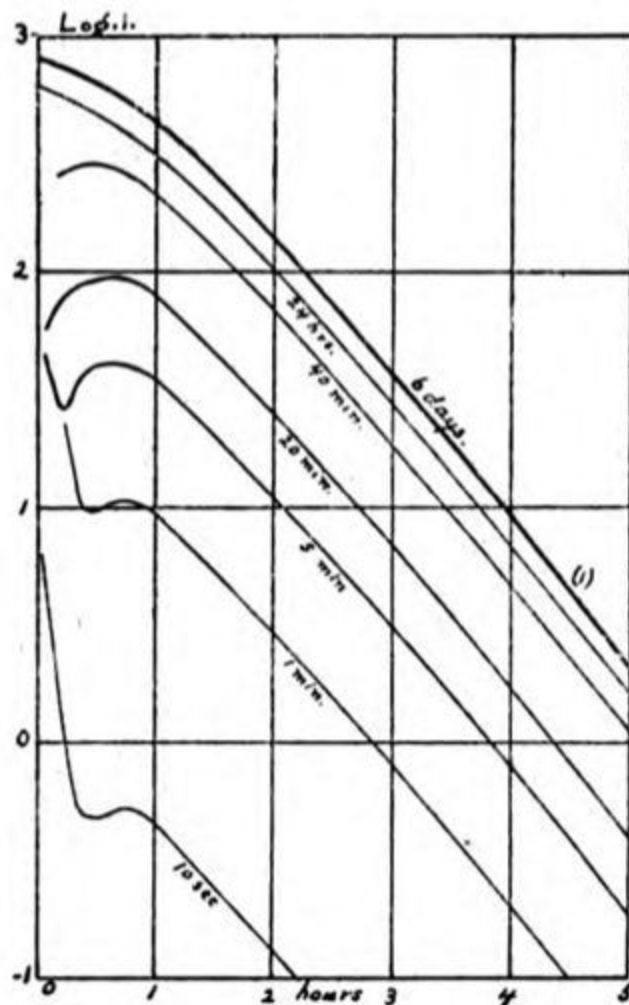


Figure V

If the duration of the action of the emanation is not so long, the law of the variation of the radiation during its loss is much more complex. In Fig. V. are represented the results of experiments for different times of action, as indicated on the corresponding curves. We see, for example, that for a time of excitation of five minutes, the intensity of the radiation during the loss of

activity first falls rapidly to a minimum; then it increases, and again begins to fall. Finally, the law of loss of activity tends towards a simple exponential law which is the same as the limiting law after prolonged excitation. These complex phenomena can be explained by assuming that on the excited plate the radioactive energy is in three distinct states, but the demonstrations relative to this subject are too long to have a place in this article.

The emanation from radium causes the energetic phosphorescence of a large number of substances. Glass reservoirs containing air charged with the emanation are luminous, Thuringian glass being the most sensitive. Phosphorescent sulphide of zinc is particularly sensitive to the action of the emanation from radium, and then gives an intense light.

If in a closed space substances become more active the greater the amount of free gas around them. When plates placed parallel to one another, and a short distance apart, are put into a closed space with radium, the faces of each plate become excited in proportion to its distance from the others. When glass tubes of different diameters are filled with the emanation and communicate with one another, the walls of those of greatest interior diameter become most active. These tubes are also most luminous. To interpret these facts it must be assumed that the air charged with the emanation acts on the walls by a radiation that arises in every part of the gaseous mass, and that the radioactivity induced upon a wall is proportional to the flow of exciting radiation received by that wall.

The Slow Evolution of Induced Activity.—A solid body acquires a very feeble, persistent, induced activity when it remains a month or less in contact with the radiation from radium. A substance withdrawn from the influence of the emanation after having been subjected to it for a long time loses its activity rapidly at first, according to the laws given. But the activity of the radiation does not disappear completely. There remains a radiation several thousand times more feeble than it was at first. This radiation is given off with extreme slowness and continues for several years. (The radiation passes through a minimum, then slowly increases for several months, but always remains very slight.)

Occlusion of the Emanation of Radium by Solids.—All solids when excited by contact with the emanation from radium acquire the property of

themselves emitting the emanation in very small quantity. They preserve this power for only twenty minutes from the time they are removed from the space containing the emanation. Nevertheless certain substances as caoutchouc, paraffin, and celluloid have the property of being saturated with the emanation, and of emitting it afterwards in abundance for several hours, or even days.

Induced Activity of Liquids.—When a liquid is placed in a closed space with radium it becomes radioactive. Water, salt solutions, petroleum, etc., can thus be excited. The liquids dissolve a certain amount of the emanation. When an excited liquid is separated from the radium and sealed up in a tube it slowly loses its activity according to the law of the destruction of the emanation (decrease to one-half in four days). When the liquid is placed in a flask open to the air it loses its activity very rapidly, and the emanation spreads into the surrounding air.

Variations of the Activity of Solutions of Radium Salts and of the Solid Salts of Radium.—When a solution of a salt of radium is exposed to the air of a room in an open vessel it becomes almost inactive. The solution emits an emanation that spreads into the room and causes induced radioactivity of the walls. The radiation of the radium is thus externalized. If the solution be enclosed in a sealed tube its activity increases little by little, and tends toward a limiting value that is reached only after several months. No doubt the emanation produced by the radium accumulates in the tube until the velocity of its spontaneous destruction is equalized by the supply from the radium.

We have seen that a salt of radium that has been freshly prepared possesses an activity which increases with the time, and becomes about five times as great as the initial activity. It seems that the emanation emitted by the radium can escape only with difficulty from the solid salt, and that it accumulates there and is transformed at once into induced radiation. An equilibrium is established when the spontaneous loss becomes sufficient to compensate the production.

When solid salt of radium is heated to redness, all the emanation which it had accumulated escapes. When the salt returns to the temperature of the room it emits Becquerel rays to a much less extent. However, the radiation recovers little by little its original value, which is reached after one or two

months. The salt which had been heated to redness no longer possesses the power of emitting the emanation, but this property may be restored to it by dissolving it and drying it at a slightly elevated temperature.

Diffusion of the Emanation from Radium.—Danne and I have studied the law of the diffusion of the emanation of radium. A large glass reservoir filled with excited air communicates with the atmosphere by means of a capillary tube. The Becquerel radiation emitted by the walls of the reservoir is measured as a function of the time, and from this is deduced the law of the escape of the emanation through the capillary tube. It is found that the rapidity of the escape of the emanation is proportional to the quantity of it in the reservoir. It varies proportionally to the cross section of the capillary tube, and inversely as its length. These laws are the same as for a gas mixed with air under the same conditions. The coefficient of the diffusion of the emanation into air is equal to 0.100 at 10°. The coefficient is, therefore, of the same order of magnitude as that of the diffusion of carbon dioxide into the air, which is equal to 0.15 at the same temperature.

Radioactivity Induced by Thorium and the Emanation of Thorium.—Thorium emits an emanation and gives rise to the phenomena of induced radioactivity. These properties have been made the subject of numerous researches by Rutherford. The action of thorium is, otherwise, considerably less intense than that of radium.

The emanation of thorium disappears spontaneously according to a simple exponential law, but its disappearance is much more rapid than that of the emanation of radium. The quantity of the emanation from thorium diminishes by one-half in about one minute and ten seconds, while in the case of radium the quantity falls to one half in four days. This considerable difference causes a profound difference in the aspect of the phenomena.

In a closed space of not too great dimensions the emanation from radium spreads almost uniformly into all parts. But under the same conditions the emanation of thorium is found to accumulate in the vicinity of the thorium, because it disappears spontaneously before it has time to diffuse any considerable distance into the air.

The radiant activity of a substance can be measured by placing it upon the lower plate of a condenser formed of two horizontal plates, and measuring the conductivity communicated to the air between the plates. If

this measurement be made with oxide of thorium, it is found that the conductivity of the air is greatly decreased when a current of air is sent between the plates. The oxide of thorium emits, indeed, an emanation that accumulates upon the substance and by its radiation helps to ionize the air between the plates. A current of air carries away the emanation as rapidly as it is set free, and the only thing left to cause ionization is the Becquerel radiation coming directly from the thorium.

If the same experiment be repeated with a salt of radium it is found that the current of air produces only a feeble effect. With uranium and polonium, which do not emit the emanation, the current of air has no effect. On the contrary, in the case of actinium, the action of the current of air is to suppress four-fifths of the conductivity of the air. It may be concluded that for thorium, and especially for actinium, the radiation of the emanation is very important in comparison with the radiation of the radioactive substance itself.

When one wishes to excite a solid to saturation with the emanation from thorium, it is necessary to cause the emanation to act for a long time, and hence new supplies of it must be brought continually to the surface of the body that is to be excited. To do this a current of air is passed through a solution of a salt of thorium and, then this current, of air charged with the emanation, is passed over the body. The solid, when excited by the emanation from thorium, loses its activity spontaneously according to an exponential law. The radiation falls to one-half every eleven hours. Hence, contrary to what takes place with the emanations, the activity induced by thorium upon solid substances disappears more slowly than that induced by radium.

Radioactivity Induced by Actinium and the Emanation from Actinium.—Actinium emits an emanation which gives a very intense radiation. This emanation disappears spontaneously with extreme rapidity, and diminishes by one-half in about one second. In air at the atmospheric pressure the emanation emitted by actinium cannot propagate itself to a greater distance than 7 or 8 mm. from the active substance, and excites bodies only when they are placed very near the source. On the contrary, in a vacuum the diffusion is rapid, and a body placed 10 cm. away from the actinium can be excited. The radioactivity induced by the actinium upon solids disappears

according to an exponential law. It diminishes to one-half in about thirty-six minutes.

The Concentration of Induced Radioactivity upon Bodies Charged Negatively.—Rutherford showed that a body exposed to the emanation from thorium became much more active when it was charged negatively than when it was at the same potential as the surrounding objects. On the contrary, it became less excited when it had a positive potential. The same phenomenon is noticed in the case of the excitation by radium and actinium. The nature of this curious phenomenon seems to me not to be well established.

Condensation of the Emanations from Radium and Thorium.—Rutherford and Soddy discovered that the emanations from radium and thorium can be condensed at the temperature of liquid air. A current of air charged with the emanation loses its radioactive properties when passing through a coil of tube plunged in liquid air. The emanations remain condensed in the tube, and can be restored to the gaseous state when it is warmed. The emanation from radium condenses at -150° , while that from thorium condenses between -100° and -150° . The experiment can be performed as follows: Two glass reservoirs, one large and the other small, communicate with one another. They are filled with gas excited by radium. The small reservoir is plunged in liquid air. The large reservoir then rapidly becomes inactive, while all the activity is concentrated in the small reservoir. If the two are then disconnected, and the small one be taken from the liquid air, it is seen that the large reservoir is not at all luminous, while the small one gives off more light than at the beginning of the experiment. The experiment is a very brilliant one if the walls of the reservoirs are coated inside with phosphorescent sulphide of zinc.

When a platinum wire excited by thorium or radium is heated to redness, it loses most of its activity. Fanny Cook Gates showed that this radioactivity is transferred to cold solids placed in the vicinity of the wire. It distils in some way, at a sufficiently high temperature, passing through the intermediate form of a gaseous emanation. The induced radioactivity of solids would thus be analogous to a condensed emanation.

Activity Induced by Remaining in Solution in a Radioactive Liquid. Uranium X. Thorium X.—Certain substances are temporarily excited when

they stay in the same solution with radioactive substances. Giesel and Mme. Curie prepared active bismuth by dissolving one of its salts in the solution of a salt of radium. Debierne also excited a salt of barium in a solution of a salt of actinium. The barium salt thus excited presents certain analogies to the salt of radium, and can be fractionated in the same way. By crystallization the active chloride is concentrated in the salt that is deposited.

By various methods of chemical precipitation the activity of uranium has been divided (Crookes, Soddy, Rutherford and Grier, Debierne, Becquerel). For instance, barium chloride is added to a solution of uranyl nitrate, and the barium is then precipitated by adding a little sulphuric acid. The precipitated barium sulphate, when separated and dried, is radioactive. It has carried with it a part of the activity of the uranium, for the uranium salt, when the solution is evaporated to dryness, is less active than before it was submitted to this operation. But after several months the sulphate of barium loses its radioactivity, while the uranium salt has regained its original properties. It is evident that the barium was excited by contact with the uranium, or that it occluded in some special form a part of the activity of the uranium (uranium X of Crookes).

Rutherford and Soddy showed that if nitrate of thorium is precipitated by ammonia, the precipitated oxide of thorium is less active than ordinary thorium. The liquid from which it was precipitated is radioactive, and on evaporating it to dryness the small residue is 2,500 times more active than the thoria (they call the radioactive substance in this residue thorium X). After several weeks the residue has lost its activity, the thorium X has disappeared, and the thoria which was precipitated has, on the contrary, regained its normal activity. Further, while the thorium X existed it emitted the thorium emanation in abundance.

Rutherford and Soddy think that uranium X and thorium X are simply intermediate products of the breaking down of uranium and thorium. Thorium, for example, produces thorium X continuously, which breaks down into the emanation of thorium, that is in its turn transformed into induced activity.

Conductivity of Atmospheric Air. Emanation and Induced Radioactivity at the Surface of the Earth—Elster and Geitel, and also Wilson, showed that

atmospheric air conducts electricity to a slight extent, that it is always slightly ionized. This ionization seems to be due to various causes. According to the work of Elster and Geitel, atmospheric air always contains a small proportion of an emanation analogous to that emitted by radioactive substances. Metallic wires suspended in the air and kept at a high negative potential become active under the influence of this emanation. At the summits of high mountains the atmosphere contains more of the emanation than on plains or at the level of the sea. The air of caves and caverns is especially highly charged with the emanation. Air rich in the emanation can also be obtained by aspirating, by means of a tube sunk in the earth the air which is contained in it. The air extracted from certain mineral waters contains the emanation, while the air dissolved in the water of the sea and of the rivers is almost free from it.

The conductivity of the atmosphere is also probably due in part to very penetrating radiations that traverse space, and of which the origin is unknown. Finally, it is probable that all bodies are slightly radioactive, and that those at the surface of the ground render the air around them a conductor of electricity.

Time-Constants that Characterize the Disappearance of Emanations and of Induced Radioactivity.—We have seen that the radioactive emanations and the induced radioactivity of solids disappear spontaneously, and that the law of their disappearance is, in general, a simple exponential law. The intensity of the radiation, I , is given as a function of the time, t , by the formula,

I_0 is the intensity of the initial radiation, and k a constant. This exponential law is completely defined by the knowledge of a constant of time that may be, for example, the inverse of k in the preceding formula. One may also take as the constant the time necessary for the intensity of the radiation to diminish to one-half.

It is very remarkable that these constants of time seem to be invariable under widely different circumstances. Thus it is that the emanation from

radium diminishes to one-half during each period of four hours, whatever may be the conditions of the experiment, and whatever the temperature between -180° and $+450^{\circ}$. The rate of its disappearance is the same whether the emanation be in the gaseous state (room temperature) or condensed (-180°). The properties of the emanation from radium thus give us an invariable standard of time which is independent of all agreements as to the unit.

The time-constants of the radioactivity serve to characterize in a precise manner the nature of the different radioactive energies.

The following are the times necessary for the activity to fall to one-half of its value:

Emanation of	Time.
Radium	4 days
Thorium	1 minute 10 seconds
Actinium	Several seconds
Induced radioactivity of	
	1 hour (after beginning of the loss).
Radium	28 minutes (for times more than two hours after the beginning of the loss)
Thorium	11 hours
Actinium	36 minutes

Thus J. J. Thomson and Adam found recently that the emanation from water from certain sources falls to one-half for each period of four days, and that this emanation causes induced activity in solids which falls to one-half in about forty minutes. The supposition is that the emanation contained in the water is due to radium.

Ordinary thorium extracted from monazite sand is slightly radioactive. Thorium from pitchblende is strongly radioactive (thorium with actinium of

Debierne). The radioactivity in the two cases is not due to the same substance, for the time-constants of the emanation and of the radioactivity are different.

Certain radioactive substances like actinium have never been separated in a pure state, and it may be supposed that the very active substances that have been studied contain only traces of them. The chemical reactions of substances cannot be recognized with certainty when they are found solely in a diluted condition, mixed with other substances. Under those circumstances one element may carry another down with it in a precipitation, and the action of the reagent is not the same as when we have pure compounds. Hence the chemical reactions will not serve to characterize a radioactive substance. It may, however, be recognized under all circumstances by the time constants of the emanation it emits, and of the induced radioactivity excited by that emanation upon solids.

Nature of the Emanation.—According to Rutherford, the emanation of a radioactive substance is a radioactive, material gas which escapes from it. In fact, the emanation from radium acts in many ways like a gas.

When we put in communication two glass reservoirs, one containing the emanation and the other none, the emanation diffuses into the second and, when equilibrium is established, it is found that the emanation is divided between the two reservoirs according to their respective volumes. One of the two may be heated to 350° , while the other remains at the room temperature, and it is found that in this case, also, the emanation is divided between the two reservoirs as if it were a perfect gas obeying the laws of Marriotte and Gay-Lussac.

We have also seen that the emanation from radium diffuses into the air according to the law of the diffusion of gases, and with a coefficient of diffusion comparable with that of carbon dioxide. Finally, the emanations from radium and thorium condense at low temperatures like liquefiable gases.

At the same time it should be remembered that no one has yet observed any pressure due to the emanation, nor has any one shown by weighing that a material gas is present. All our knowledge of the properties of the emanation results from measurements of radioactivity. More than that, no

one has yet shown with certitude that there is a characteristic spectrum produced by the emanation.

The emanation should also not be considered as an ordinary material gas, for it disappears spontaneously from a sealed tube containing it, and the rapidity of its disappearance is absolutely independent of the conditions of the experiment, especially of the temperature.

It is very curious that the numerous attempts made under the most varied conditions to obtain chemical reactions with the emanation have been fruitless. To explain this fact Rutherford thinks that the emanations are gases of the argon family.

The following facts are difficult to explain. The emanation from radium condenses at -150° , and according to Rutherford, a current of air at -153° may be passed over it continuously without removing it. The amount of the emanation must be very small, and if it had the slightest tension at -153° , it would quickly evaporate in a current of air. Further, the temperature of condensation by cooling should be a function of the amount of the emanation in a given volume of air, which has not yet been proved.

Debierne and I have found that the emanation passes with extreme ease through the tiniest holes and fissures in solids, while under the same conditions the ordinary gases could circulate only with the greatest slowness.

Rutherford supposes that radium destroys itself spontaneously, and that the emanation is one of the products of the breaking-down. Debierne and I observed that a solid salt of radium quite rapidly excites the walls of the reservoir filled with air, which contains it, by the emanation which it emits. On the contrary, if a quite perfect vacuum be made in the reservoir the excitation takes place only with extreme slowness; but it rapidly reappears when a gas has been admitted. However, the emanation spreads much more rapidly in a gas at very low pressure than at the atmospheric pressure. It seems from this that the emanation experiences some particular difficulty in escaping from radium which is in a vacuum.

Disengagement of Gas by the Salts of Radium. Production of Helium.—Giesel noticed that solutions of radium bromide continually give off gases. These gases are principally hydrogen and oxygen, in the same relative

proportion as in water, and might, therefore, come from the decomposition of the water of the solution. But Ramsay and Soddy also showed that there always is a small quantity of helium that they detected by its spectrum in a Geissler tube. The helium lines were also accompanied by three unknown lines.

A solid salt of radium also constantly gives off gases capable of producing a pressure in a closed tube. To this liberation of gas can be attributed two accidents during my experiments. A sealed bulb of thin glass, almost filled with well-dried bromide of radium, exploded and became slightly warm at the same time. An explosion was also caused by dry radium chloride when heated quite rapidly in a vacuum to 300° . In this case the explosion seemed to be caused by fragments of the solid salt filled with occluded gas.

At the moment we dissolve in water a solid salt of radium that has been prepared a long time, there is an abundant evolution of gas.

The spontaneous production of helium in a sealed tube containing radium is plainly a new fact of fundamental importance. Ramsay and Soddy accumulated some of the emanation from radium, and enclosed it in a Geissler tube at low pressure. They obtained new lines which they attributed to the emanation, and they also showed that the spectrum of helium was absent at first, but that it came into being little by little in the tube. According to that, helium must be one of the products of the disintegration of radium.

In support of the preceding results may be mentioned some points noticed by Mme. Curie and me at the beginning of our work. We were struck by the simultaneous occurrence of uranium, radium, and helium in the same mineral. We took 50 kilograms of commercial barium chloride, coming from minerals that did not contain uranium, and submitted it to fractional crystallization to see whether it contained traces of radium chloride. After a prolonged fractionation, the portion at the head, now reduced to a few grams, was not at all radioactive. Hence, barium contains radium only when it comes from uranium minerals. These are the same that contain helium. One might think that there is a relation of cause and effect from the simultaneous occurrence of these three substances.

This rapid summary of the researches on radioactivity serves to show the importance of the scientific movement that has been started by the study of this phenomenon. The results obtained are of a nature to modify the ideas one might have about the invariability of the atom, the conservation of matter and of energy, the nature of the mass of bodies, and the energy spread through space. The most fundamental questions of science are thus brought into the discussion. Apart from the theoretical interest of which they are the object, the phenomena of radioactivity give new means of action to the physicist, the chemist, the physiologist, and the physician.

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